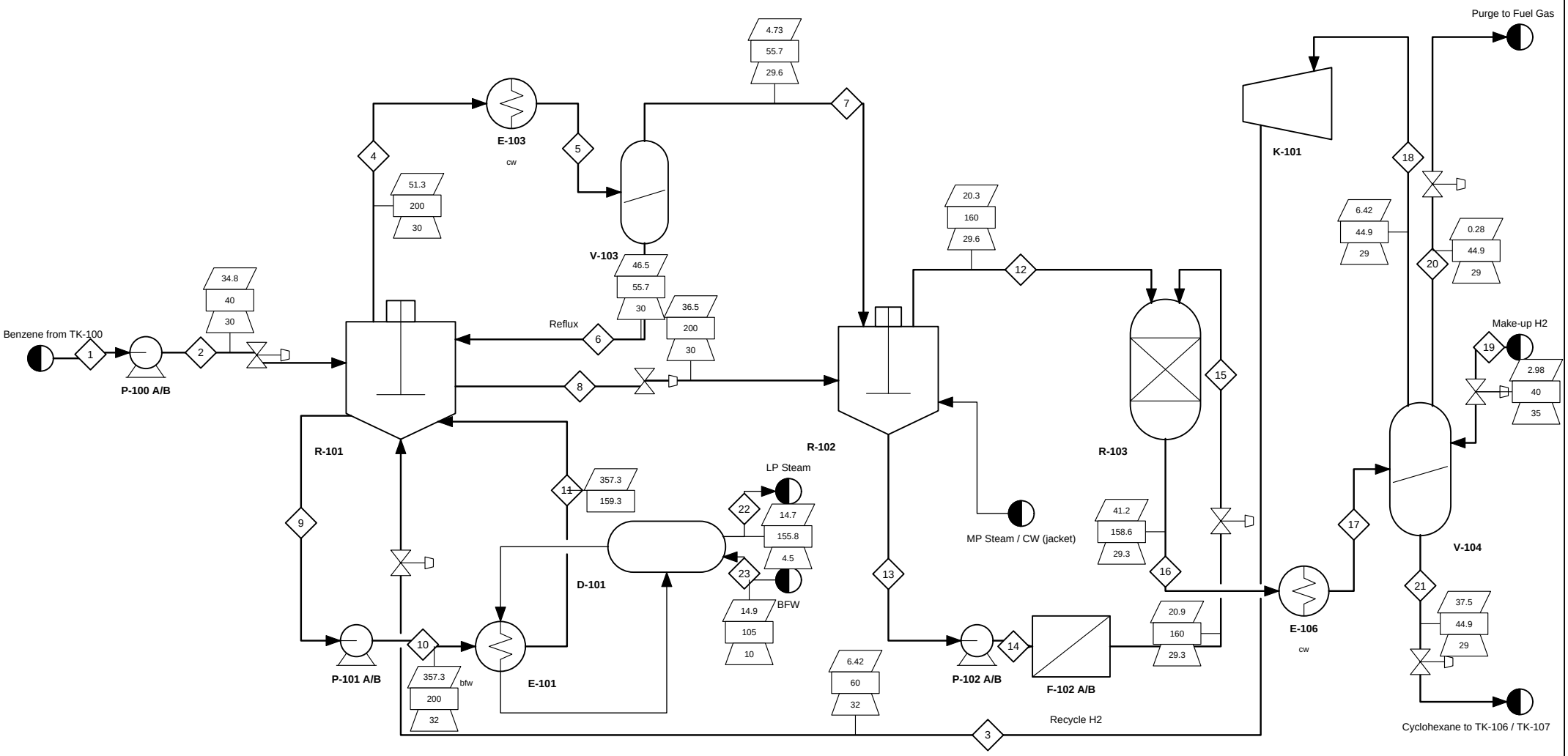
P-100 A/B	R-101	E-103	V-103	P-101 A/B	E-101	D-101	R-102	P-102 A/B	F-102 A/B	R-103	E-106	V-104	K-101
Benzene Feed Pumps	Slurry Reactor + Agitator M-101	Reflux Condenser	Reflux Drum	Pump-around Pumps	Pump-around Cooler / LP Steam Generator	Steam Drum	Slurry Finishing Reactor (jacketed)	R-102 Transfer Pumps	Catalyst Candle Filters	Trickle-bed Finishing Reactor	Product Cooler	HP Separator	Recycle H2 Compressor



REACTION

C6H6 + 3 H2 -> C6H12 dH = -206 kJ/mol

Raney-Ni slurry (R-101/102), Ni bed (R-103)

Benzene 34.8 t/h; product 37.5 t/h, 2 ppm benzene

NOTES

1. Rev 0 · Issued for teaching. Flags: t/h, °C, bar g at the design point.
2. All numbers come from the simulator model (design state, noise off); no flag = no model value.
3. H2 passes R-101, E-103/V-103, R-102, R-103, V-104 and K-101 in series; D-101 blowdown (≈1 %) not shown.

LEGEND

Mass Flow (t/h) · Temperature (°C) · Pressure (bar g) · Process Line · Utility Line · Control Valve

Process Input / Output · Stream Number · Line Crossing · Centrifugal Pump · Stirred Reactor · Heat Exchanger

Separator · Vessel / Drum · Filter · Packed-Bed Reactor · Compressor

CYCLOHEXANE REACTOR VER 1 · OPERATOR TRAINING SIMULATOR

PROCESS FLOW DIAGRAM

CYCLOHEXANE UNIT — BENZENE HYDROGENATION REACTOR TRAIN

DRAWING No.	REV.	SIZE	SHEET
CHX-V1-PFD-001	0	A4	1 OF 1
DRAWN	**CHECKED**	**SCALE**	**DATE**
Apinan Soottitantawat		NTS	2026-10-26